

1 Interface QM-IDI

1 Introduction

QM-IDI is an interface designed for exchanging inspection processing data between the QM application component and external systems. Inspection specifications defined in the QM component are transferred to the external system. The external system independently carries out the inspection and transfers the results back to QM.

The interface is implemented using synchronous Remote Function Calls (sRFC), whereby all communication is started from HYDRA. The documentation at hand describes the processing of data exchange between SAP R/3 QM and HYDRA and the different types of integration with other SAP standard interface such as PP-PDC.

2 SAP interface technologies

2.1 RFC (Remote function call)

The remote function call is the basis technology for the system-wide call of programs in R/3 or its partner systems. RFC is triggered by the client and transmitted to the server of the partner system. There are the following variants:

Synchronous RFC (sRFC), the calling program waits until the function module that was called is processed and the results are transmitted.

Asynchronous RFC (aRFC), it is established whether the system being called is available, but execution of the function is not awaited.

Transactional RFC (tRFC), the system which is called guarantees execution of all function modules, or in case of error, to change nothing. tRFC also runs asynchronously.

PP-PDC only uses this type of remote function call.

2.2 BAPI (Business application programming interface)

Under the term BAPI, SAP provides a range of ready-made interfaces, which give partner systems access to the functionality of the R/3 system. (Communication is technically based on RFC.)

BAPIs enable the integration of systems on a business level!

2.3 ALE (Application link enabling)

ALE is a technology which is used for the assembly and operation of distributed applications (R/3, R/2 and third-party systems). ALE offers a controlled exchange of business messages and data and consistent data management for loosely coupled (asynchronous) systems.

2.4 IDoc (Intermediate document)

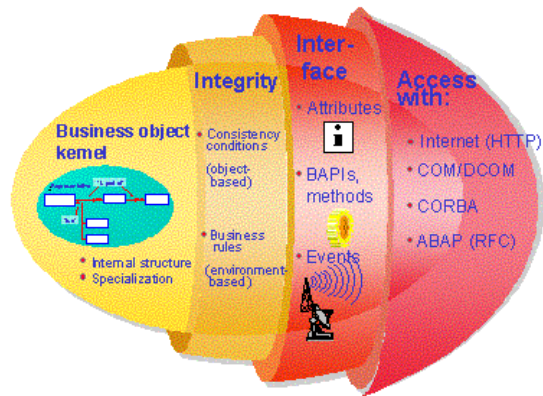
In the SAP environment IDocs are containers for data exchange between systems. IDocs can either be flat or they can form multi-level hierarchies.

Every IDoc comprises the following components:

Control record, contains information on IDoc type, headers, sender and receiver

Data record, the actual data is found here as the segments of a flat ASCII file.

Status record, contains information on the status of the IDOC, e.g. error messages.



Kernel:

innermost layer which contains business object data and structure

Integrity layer:

contains the business logic of the business object

Interface layer:

enables communication with distributed systems

Access layer:

defines the technologies allowed to access the object data, e.g. tRFC

Figure: Layer model of SAP business objects (from SAP online help)

3 Mapping the QM-IDI in HYDRA

3.1 Features of the QM-IDI

In the context of a connection via the QM-IDI interface, it is the task of HYDRA to receive inspection processing data and to integrate them into the HYDRA process. In addition to that, HYDRA creates the appropriate data from recorded confirmations and transfers these to R/3. Both scenarios are initiated by HYDRA.

3.2 R/3 → HYDRA – movement and master data

3.2.1 Download of inspection data

The transfer of inspection batch data from SAP to HYDRA will be carried out using the function module QIRF_SEND_REQUIREMENTS_GET_DAT2. For that purpose, HYDRA transfers selection options to the function module. Those options are transferred using the structure QAILS. Additionally, the function module offers the possibility to control the download in detail. The content of each parameter can be configured in HYDRA in a customizing table (please see chapter 6.2 *Configuring function module parameters* for more details).

Parameter name:	T	L	D	Description
I_IND_SORT_ASCENDING_TO_DATE	CHAR	1		Indicator: Sort operations in ascending order according to creation date Default = "X"
I_IND_SORT_DESCENDING_TO_DATE	CHAR	1		Indicator: Sort operations in descending order according to creation date Default= " "
I_IND_MULTI_TRANSFER_POSSIBLE	CHAR	1		Indicator: Correction transfers of operations possible, Default = " "
I_IND_EVALUATE_CHARACTERISTIC	CHAR	1		Indicator: Transfer valuation specifications for characteristics, Default = "X"
I_IND_SET_BLOCK_INDICATORS	CHAR	1		Indicator: Set lock entries in QM, Default = "X"
I_IND_ONLY_OBLIGATORY_CHARACT	CHAR	1		Indicator: Only transfer required characteristics, Default = " "

Parameter name:	T	L	D	Description
I_IND_TRANSFER_CHAR_CODES	CHAR	1		Indicator: Transfer catalog data for inspection characteristics, Default = "X"
I_IND_READ_WORK_CENTER	CHAR	1		Indicator: Read work center data, Default = "X"
I_IND_READ_VENDOR_AND_PRODUCER	CHAR	1		Indicator: Read vendor data, Default = "X"
I_IND_READ_PURCHASING_INFO				Indicator: Read data from purchasing information record, Default = "X"
I_IND_READ_SALES_INFO				Indicator: Read data from sales information record, Default = "X"
I_IND_SEND_PROTOCOL_MAIL				Indicator: Send error log by mail Default = "X"
I_IND_TRANSFER_USAGE_DEC_CODES				Indicator: Transfer catalog data for usage decision, Default = "X"

The structure QAILS contains the selection options. The content depends on the business scenario and will be described separately. It has the following structure:

Field	T	L	Description
SATZART	CHAR	3	Record type for request record
LOSNR_VON	NUMC	12	From inspection batch number
LOSNR_BIS	NUMC	12	To inspection batch number
PLNFL	CHAR	6	Operation sequence in task list
VORNr_VON	CHAR	4	From operation number
VORNr_BIS	CHAR	4	To operation number
VORGWERK	CHAR	4	Plant of operation to be processed
SUBSYS	CHAR	6	Identifier of the subsystem
PRPLATZ	CHAR	8	Work center
PRPLATZWRK	CHAR	4	Plant of the work center
MATNR	CHAR	18	Material number
DATUM_VON	DATE	8	From creation date of inspection batch
DATUM_BIS	DATE	8	To creation date of inspection batch
PRUEFSTAT	CHAR	1	Status of the inspection
ART	CHAR	8	Inspection type
HERKUNFT	CHAR	2	Origin of the inspection batch
CHARG	CHAR	10	Batch number
AUFNR_VON	CHAR	12	From order number

Field	T	L	Description
AUFNR_BIS	CHAR	12	To order number
LIFNR	CHAR	10	Vendor number
KUNNR	CHAR	10	Customer number

According to the selection option, the function module provides the inspection batch data, including inspection batch header, operation and characteristics. The structures will be explained in detail in chapter 4 *Download of inspection specifications / master data*, also pointing out the meaning of the different fields.

Besides the inspection data, the function module also provides a table of error messages. These error messages will be displayed in the MLE inbound transactions and in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license HYD-ESK and HKM-ESK) with escalation SAP.QM_IDI_INBOUND_MSG.

The download of inspection batch data will be carried out by the hysapqmc.exe/out program.

3.2.2 Download catalog master data

The transfer of catalog master data from SAP to HYDRA will be carried out by using the function module QIRF_SEND_CATALOG_DATA2. For that purpose, HYDRA transfers selection options to the function module. The content of each parameter can be configured in HYDRA in a customizing table (please see chapter 6.2 *Configuring function module parameters* for more details).

Parameter name:	T	L	D	Description
I_IND_CATALOG_IS_SEL_SET	CHAR	1		Indicator: Choose selected set, Default = " "
I_IND_CATALOG_IS_CODEGROUP	CHAR	1		Indicator: Choose code groups Default = " "
I_IND_CATALOG_TYPE	CHAR	3		Catalog type Default = "3"
I_IND_PLANT_OF_SELECTED_SET	CHAR	4		Plant of selected set Default = "0001"
I_IND_CATALOG	CHAR	4		Catalog (selected set or code group)
I_IND_LANGUAGE	CHAR	2		Language Default = "D"

Besides the catalog data, the function module also provides a table of error messages. These error messages will be displayed in the MLE inbound transactions, the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license HYD-ESK and HKM-ESK) with escalation SAP.QM_IDI_INBOUND_MSG.

The download of catalog master data will be carried out by the hysapqmc.exe/out program.

3.2.3 Download inspection point data

The transfer of inspection point data from SAP to HYDRA will be carried out by using the function module QIRF_INSPPOINT_GETLIST. For that purpose, HYDRA transfers selection options to the function module. The content of each parameter can be configured in HYDRA in a customizing table (please see chapter 6.2 *Configuring function module parameters* for more details).

Parameter name:	T	L	D	Description
INSPLOT	-	-		No usage
INSPOPER	CHAR	4		
INSPPOINT_FROM	NUMC	6		Inspection point from Default = "000001"
INSPPOINT_TO	NUMC	6		Inspection point to Default = "999999"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“

Besides the inspection point data, the function module also provides a table of error messages. These error messages will be displayed in the MLE inbound transactions, the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license HYD-ESK and HKM-ESK) with escalation SAP.QM_IDI_INBOUND_MSG.

The download of inspection point master data will be carried out by the hysapqmc.exe/out program.

3.3 HYDRA → R/3 – inspection results / inspection points

HYDRA records results according to the inspection specifications. Those are to be transferred to SAP QM. For that purpose, SAP provides a set of function modules. According to the record type of the recorded inspection results one or the other has to be used.

3.3.1 Upload for Single Results

The QIRF_GET_ORIGINAL_VALUES2 function module has to be used to upload single results for inspections. Besides the QAISE structure, which carries the actual inspection results, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in a customizing table (please see chapter 6.2 *Configuring function module parameters* for more details).

Parameter name:	T	L	D	Description
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Parameter name:	T	L	D	Description
I_IND_EVALUATION_TRANSFER	CHAR	1		Indicator: Transfer valuations from subsystem Default = "X"
I_IND_CLOSE_PROCESSING	CHAR	1		Indicator: Close samples or characteristics, Default = " "
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“
I_IND_POSTING_KZ	CHAR	1		Indicator: Call up program for updating results, Default = "X"
I_IND_PROC_COMMIT_WORK	CHAR	1		Indicator.: Trigger commit work to database, Default = "X"

As a result of the posting process in SAP QM, the function module provides a table of error messages. These messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license HYD-ESK and HKM-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

3.3.2 Upload for Sample Results

The QIRF_GET_SAMPLE_VALUES2 function module has to be used to upload sample results for inspections. Besides the QAISR structure, which carries the actual inspection results, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in a customizing table (please see chapter 6.2 *Configuring function module parameters* for more details).

Parameter name:	T	L	D	Description
I_IND_EVALUATION_TRANSFER	CHAR	1		Indicator: Transfer valuations from subsystem Default = "X"
I_IND_CLOSE_PROCESSING	CHAR	1		Indicator: Close samples or characteristics, Default = " "
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“
I_IND_POSTING_KZ	CHAR	1		Indicator: Call up program for updating results, Default = "X"

Parameter name:	T	L	D	Description
I_IND_PROC_COMMIT_WORK	CHAR	1		Indicator.: Trigger commit work to data-base, Default = "X"

As a result of the posting process in SAP QM, the function module provides a table of error messages. These error messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license HYD-ESK and HKM-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

3.3.3 Upload for Characteristic Results

The QIRF_GET_FEATURE_VALUES2 function module has to be used to upload characteristic results for inspections. Besides the QAIMR structure, which carries the actual inspection results, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in a customizing table (please see chapter 6.2 *Configuring function module parameters* for more details).

Parameter name:	T	L	D	Description
I_IND_EVALUATION_TRANSFER	CHAR	1		Indicator: Transfer valuations from subsystem Default = "X"
I_IND_CLOSE_PROCESSING	CHAR	1		Indicator: Close samples or characteristics, Default = " "
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“
I_IND_POSTING_KZ	CHAR	1		Indicator: Call up program for updating results, Default = "X"
I_IND_PROC_COMMIT_WORK	CHAR	1		Indicator.: Trigger commit work to data-base, Default = "X"

As a result of the posting process in SAP QM, the function module provides a table of error messages. These error messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license HYD-ESK and HKM-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

3.3.4 Upload for Inspection Points

Creating and updating / deleting inspection points are done by using the QIRF_GET_INSP_POINT2 function module. Besides the QAIPP structure, which carries the actual inspection point data, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in a customizing table (please see chapter 6.2 *Configuring function module parameters* for more details).

Parameter name:	T	L	D	Description
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“

As a result of the posting process in SAP QM, the function module provides a table of error messages. These error messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license HYD-ESK and HKM-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

3.3.5 Upload for Usage Decision

The usage decision is uploaded by using the QIRF_GET_USAGE_DECISION2 function module. Besides the QAIVE structure, which carries the actual usage decision data, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in a customizing table (please see chapter 6.2 *Configuring function module parameters* for more details).

Parameter name:	T	L	D	Description
I_STOCK_POSTING	CHAR	1		Indicator: Trigger inventory posting in QM after usage decision Default = "X"
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“

As a result of the posting process in SAP QM, the function module provides a table of error messages. These error messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license HYD-ESK and HKM-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

3.3.6 Upload for Defect Items

The defect items are uploaded by using the QIRF_GET_DEFECT_ITEMS2 function module. Besides the QMIFE structure, which carries the actual defect items data, the function module offers additional parameters that allow controlling the posting process in SAP QM. The content of each parameter can be configured in HYDRA in a customizing table (please see chapter 6.2 *Configuring function module parameters* for more details).

Parameter name:	T	L	D	Description
I_SEND_PROTOCOL_MAIL	CHAR	1		Indicator: Send error log by mail, Default = "X"
I_SUBSYS	CHAR	6		Subsystem confirms results Default = „QM0001“

As a result of the posting process in SAP QM, the function module provides a table of error messages. These error messages will be displayed in the MLE outbound transactions in the protocol of the communication program. Additionally, those entries are also forwarded to the HYDRA Escalation Management (Prerequisite: license HYD-ESK and HKM-ESK) with escalation SAP.QM_IDI_OUTBOUND_MSG.

3.4 Supported integration scenarios

Each function module provided on the SAP side expects specific import parameters, structures or tables. Depending on the function module, the data in these ones are derived either from configuration in HYDRA or from integration with other interfaces. The implementation of the QM-IDI in HYDRA serves the following processes and scenarios:

- Integration with SAP standard interface PP-PDC
- Integration with SAP standard interface PP-PDC and HYDRA interface extension HYINFO
- Stand-alone CAQ (e.g. for goods receipt)

3.4.1 Integration with SAP standard interface PP-PDC

The integration between QM-IDI and PP-PDC interface serves the requirements of increased in process control during production process. For that reason the routing in SAP not only contains productive operation (transferred via PP-PDC interface) but also QM operations that itself contain characteristics and which are transferred using QM-IDI interface. It is also possible that an operation is transferred via both interfaces.

The download process is started by releasing the production order in SAP. That causes a download of the production order and its operations via PP-PDC interface. The data will be received in HYDRA in the MES Link Enabling layer for further processing. Instead of performing the posting into the HYDRA application tables, the data is used to receive the inspection data. For that specific data of the PP-PDC IDoc E2BP-PP-PDC-OPERA2000 is taken and transferred into the QM-IDI QAILS structure to request the inspection batch for each order. The following table indicates which fields of the PP-PDC IDoc are used to request inspection specifications:

Field	Type	L	Description	Usage / Origins
SATZART	CHAR	3	Record type to request record	Fix „Q40“
LOSNR_VON	NUMC	12	From inspection batch number	Not used
LOSNR_BIS	NUMC	12	To inspection batch number	Not used
PLNFL	CHAR	6	Operation sequence in task list	Not used
VORNR_VON	CHAR	4	From operation number	Not used
VORNR_BIS	CHAR	4	To operation number	Not used
VORGWERK	CHAR	4	Plant of operation to be processed	E2BP_PP_PDC_OPERA2000. PLANT
SUBSYS	CHAR	6	Identifier of the subsystem	Customizing table in HYDRA
PRPLATZ	CHAR	8	Work center	Not used
PRPLATZWRK	CHAR	4	Plant of the work center	Not used
MATNR	CHAR	18	Material number	E2BP_PP_PDC_OPERA2000. MATERIAL
DATUM_VON	DATE	8	From creation date of inspection batch	Not used
DATUM_BIS	DATE	8	To creation date of inspection batch	Not used
PRUEFSTAT	CHAR	1	Status of the inspection	Not used
ART	CHAR	8	Inspection type	Not used
HERKUNFT	CHAR	2	Origin of the inspection batch	Not used
CHARG	CHAR	10	Batch number	Not used
AUFNR_VON	CHAR	12	From order number	E2BP_PP_PDC_OPERA2000. ORDERID
AUFNR_BIS	CHAR	12	To order number	E2BP_PP_PDC_OPERA2000. ORDERID

LIFNR	CHAR	10	Vendor number	Not used
KUNNR	CHAR	10	Customer number	Not used

As a result of the download of the inspection specification the original PP-PDC IDoc and the newly transferred QM-IDI data are pooled into a new IDoc of the following characteristics:

Message type / IDoc type: PPCC2RECORDER_QM_IDI

Segments:	E2PPCC2RECORDER (Indicator initial download)	1
	E2BP_PP_PDC_OPERA2000 (PP operations)	1-n
	E2BP_PP_PDC_OPERA1000 (Deletion of PP operations.)	0-n
	Z2QIAVC000X000 (Inspection batches and operations)	1-n
	Z2QAIMV000X000 (inspection characteristics)	1-n
	Z2QAICA000X000 (catalogs)	1-n
	Z2QIERR000X000 (error logs)	1-n

Prerequisite:

- HYDRA license HKMPP-PDC

3.4.2 Integration with SAP PP-PDC and HYINFO

The integration of QM-IDI with PP-PDC can also be extended when using HYDRA interface extension for PP-PDC HYINFO. The interface extension HYINFO also uses sRFC calls to receive additional data from SAP such as order header data or the list of components.

In this case, the original PP-PDC will be used first to collect additional data within SAP from the interface extension. As a result the interface extension pools the original order data from PP-PDC interface and the newly selected additional data into a new IDoc of the following characteristics:

Message type / IDoc type: PPCC2HYINFOORDER

Segments:	E2PPCC2RECORDER (Indicator initial download)	1
	E2BP_PP_PDC_OPERA2000 (PP operations)	1-n
	E2BP_PP_PDC_OPERA1000 (Deletion of PP operations.)	0-n
	HYINFO_xxx (additional data from interface extension)	1-n

This new IDoc is the basement to collect inspection specifications in SAP QM. For that, the original PP-PDC data will be used to derive the production orders to request the specifications for. For that the same data / fields will be used as described in chapter 3.4.1 *Integration with SAP standard interface PP-PDC*. As a result a new IDoc will be created by pooling as much the original PP-PDC data, the data from interface extension HYINFO as the QM-IDI data with the following characteristics:

Message type / IDoc type: PPCC2HYINFOORDER_QM_IDI

Segments:	E2PPCC2RECORDER (Indicator initial download)	1
	E2BP_PP_PDC_OPERA2000 (PP operations)	1-n
	E2BP_PP_PDC_OPERA1000 (Deletion of PP operations.)	0-n
	HYINFO_xxx (additional data from interface extension)	1-n
	Z2QIAVC000X000 (Inspection batches and operations)	1-n
	Z2QAIMV000X000 (inspection characteristics)	1-n
	Z2QAICA000X000 (catalogs)	1-n
	Z2QIERR000X000 (error logs)	1-n

3.4.3 Stand-alone CAQ

The stand-alone variant is thought to support inspections that are not related to production orders, such as goods receipt inspections. It can be used independently or in combination with the two integration scenarios described in the previous chapters.

In order to be able to request the inspection batches for those inspections, there is a customizing table in HYDRA that allows defining different selection options. To support different kinds of selection options it is possible to configure multiple variants. The details of the customizing table are described in chapter 6.1 *Configuring variants for Stand-alone CAQ*.

The request of inspection batch will be performed on a cyclic base. For the calling program hysapqmc.exe/out has to be configured in the HYDRA Scheduler.

As a result of the receipt of inspection batches an IDoc will be created with the following characteristics:

Message type / IDoc type: ZQM_IDI

Segments:	Z2QIAVC000X000 (Inspection batches and operations)	1-n
	Z2QAIMV000X000 (inspection characteristics)	1-n
	Z2QAICA000X000 (catalogs)	1-n
	Z2QIERR000X000 (error logs)	1-n

4 Download of inspection specifications / master data

The processing of all other segments in the integration scenario with PP-PDC and / or HYINFO interface extension are described in their respective interface documentations (HKMPP-PDC.pdf or HKMPP-INF.pdf). In this chapter the QM-IDI processing is described only.

4.1 Inspection batch / operations in “Z2QAIVC000X000” structure

Field name:	T	L	D	Description	Use in HYDRA
SATZART	CHAR	3		Record type for header record	Not used
PRUEFLOS	NUMC	12		Number of the inspection batch	Number of the inspection batch
PLNFL	CHAR	6		Operation sequence in task list	In accordance with configuration (*1)
VORNR	CHAR	4		Operation number	In accordance with configuration (*1) Display in HYDRA client (*2)
WERK	CHAR	4		Plant of the inspection batch	Plant of the inspection batch
ART	CHAR	8		Inspection type	Inspection type
HERKUNFT	CHAR	2		Origin of the inspection batch	Origin of the inspection batch
ENTSTEHDAT	DATE	8		Creation date of the inspection batch	Creation date of the inspection batch
ERSTELLER	CHAR	12		User who created the data record	User who created the data record
AENDERER	CHAR	12		User who changed the data record	User who changed the data record
AENDERDAT	DATE	8		Change date of the data record	Change date of the data record
KZVESUBSYS	CHAR	1		Ind.: make usage decision in subsystem	“X” Usage decision will be made in HYDRA “ “ Usage decision will be made in SAP
VKATART	CHAR	1		Catalog type for usage decision	Catalog type for usage decision
VWERKS	CHAR	4		Plant of selected set for usage dec.	Plant of selected set for usage dec.
VAUSWAHLMG	CHAR	8		Selected set for usage decision	Selected set for usage decision
PPVEKATART	CHAR	1		Catalog type for inspection point valuation	Catalog type for inspection point valuation
PPVEWERK	CHAR	4		Plant of the selected set for the inspection point valuation	Plant of the selected set for the inspection point valuation
PPVEMENGE	CHAR	8		Selected set for the inspection point valuation	Selected set for the inspection point valuation
PPVECODGRA	CHAR	4		Code group proposal when inspection point is accepted (acceptance of all characteristics)	Code group proposal when inspection point is accepted (acceptance of all characteristics)
PPVECODEA	CHAR	4		Code proposal when inspection point is accepted	Code proposal when inspection point is accepted
PPVECODGRR	CHAR	4		Code group proposal when inspection point is rejected (rejection of one characteristic at least)	Code group proposal when inspection point is rejected (rejection of one characteristic at least)

Field name:	T	L	D	Description	Use in HYDRA
PPVECODER	CHAR	4		Code proposal when inspection point is rejected	Code proposal when inspection point is rejected
PLNTY	CHAR	1		Task list type	Task list type
PLNNR	CHAR	8		Key of task list group	Key of task list group
PPLVERW	CHAR	3		Task list usage	Task list usage
PLNAL	CHAR	2		Task list group counter	Task list group counter
ZAEHL	NUMC	8		Version counter for task list	Version counter for task list
PLANKTEXT	CHAR	40		Short text of the task list	Short text of the task list
DATUV	DATE	8		Start of validity for task list header	Start of validity for task list header
PASTRTERM	DATE	8		Start date of the inspection	Start date of the inspection
PAENDTERM	DATE	8		Finish date of the inspection	Finish date of the inspection
KUNNR	CHAR	10		Customer number	Customer number
NAME1KUN	CHAR	35		Name 1 of the customer	Name 1 of the customer
LIFNR	CHAR	10		Vendor number	Vendor number
NAME1LIF	CHAR	35		Name 1 of the vendor	Name 1 of the vendor
HERSTELLER	CHAR	10		Number of the manufacturer	Number of the manufacturer
NAME1HER	CHAR	35		Name 1 of the manufacturer	Name 1 of the manufacturer
MATNR	CHAR	18		Material number	Material number
KTEXTMAT	CHAR	40		Short text of the material	Short text of the material
KTEXTLOS	CHAR	40		Short text of the inspection batch	Short text of the inspection batch
CHARG	CHAR	10		Batch number	Batch number
LAGORTCHRG	CHAR	4		Storage location of the batch	Storage location of the batch
LICHN	CHAR	15		Batch number used by vendor	Batch number used by vendor
IDNLF	CHAR	35		Material number used by vendor	Material number used by vendor
KDMAT	CHAR	35		Material number used by customer	Material number used by customer
POSTX	CHAR	40		Mat. short text used by customer	Mat. short text used by customer
WERKVORG	CHAR	4		Plant of the goods movement	Plant of the goods movement
LAGORTVORG	CHAR	4		Storage location for the goods movement	Storage location for the goods movement
LOSMENGE	CHAR	17		Inspection batch quantity	Inspection batch quantity
MENGENEINH	CHAR	3		Base unit of measure of the inspection batch	Base unit of measure of the inspection batch
GESSTICHPR	CHAR	17		Sample size for inspection batch	Sample size for inspection batch
EINHPROBE	CHAR	3		Unit of measure for sample	Unit of measure for sample
EBELN	CHAR	10		Purchasing document number	Purchasing document number
EBELP	NUMC	5		Item no. of purchasing document	Item no. of purchasing document
MJAHR	NUMC	4		Year of the material document	Year of the material document
MBLNR	CHAR	10		Number of the material document	Number of the material document
ZEILE	NUMC	4		Item in material document	Item in material document
BUDAT	DATE	8		Posting date in document	Posting date in document
AUFNR	CHAR	12		Order number	Order number
KDAUF	CHAR	10		Customer order number	Customer order number
KDPOS	NUMC	6		Item number in sales order	Item number in sales order
VORKTXT	CHAR	40		Short text for operation	Short text for operation
PRPLATZ	CHAR	8		Work center	HYDRA Machine / Work center
PRPLATZWRK	CHAR	4		Plant of the target work center	Plant of the target work center
PRPLATZTXT	CHAR	40		Short text of the work center	Short text of the work center
SUBSYS	CHAR	6		Identifier of the subsystem	Identifier of the subsystem

Field name:	T	L	D	Description	Use in HYDRA
QKZPRZEIT	CHAR	1		Ind.: work cycle = time	"X" Inspection point based on time intervals
QKZPRMENG	CHAR	1		Ind.: work cycle = quantity	"X" Inspection point based on quantity intervals
QKZPRFREI	CHAR	1		Ind.: any work cycle	"X" Free inspection points
QRASTZEHT	CHAR	3		Time unit of inspection grid	Supported units: "S" → seconds "SEC" → seconds "MIN" → minutes "H" → hours "HUR" → hours "STD" → hours
QRASTZFAK	NUMC	6		Time factor for inspection grid	Value for the time interval
QRASTMENG	CHAR	17		Quantity between two inspections	Value for the quantity interval
QRASTEREH	CHAR	3		Unit of measure of the insp. grid	Unit of measure for the quantity interval
PPKTTYP	CHAR	1		Inspection point type	Type of inspection point " " IP for IPC "1" IP for equipment "2" IP functional location "3" IP for physical sample
KZEQUNR	CHAR	1		Indicator: user field EQUNR active	"X" Field is active " " Field is not active
SWEQUNR	CHAR	20		Key word for user field EQUNR	
KZTPLNR	CHAR	1		Indicator: user field TPLNR active	"X" Field is active " " Field is not active
SWTPLNR	CHAR	20		Key word for user field TPLNR	
KZPHYNR	CHAR	1		Indicator: user field PHYNR active	"X" Field is active " " Field is not active
SWPHYNR	CHAR	20		Key word for user field PHYNR	
KZUSERC1	CHAR	1		Indicator: user field USERC1 active	" " Field is not active "X" Optional field "1" Obligatory field
SWUSERC1	CHAR	20		Key word for user field USERC1	Key word displayed when creating / closing inspection points
KZUSERC2	CHAR	1		Indicator: user field USERC2 active	" " Field is not active "X" Optional field "1"..."6" Obligatory field
SWUSERC2	CHAR	20		Key word for user field USERC2	Key word displayed when creating / closing inspection points
KZUSERN1	CHAR	1		Indicator: user field USERN1 active	" " Field is not active "X" Optional field "1"..."6" Obligatory field
SWUSERN1	CHAR	20		Key word for user field USERN1	Key word displayed when creating / closing inspection points
KZUSERN2	CHAR	1		Indicator: user field USERN2 active	" " Field is not active "X" Optional field "1"..."6" Obligatory field
SWUSERN2	CHAR	20		Key word for user field USERN2	Key word displayed when creating / closing inspection points
KZUSERD1	CHAR	1		Indicator: user field USERD1 active	" " Field is not active "X" Optional field "1"..."6" Obligatory field

Field name:	T	L	D	Description	Use in HYDRA
SWUSERD1	CHAR	20		Key word for user field USERD1	Key word displayed when creating / closing inspection points
KZUSERT1	CHAR	1		Indicator: user field USERT1 active	" " Field is not active "X" Optional field "1"... "6" Obligatory field
SWUSERT1	CHAR	20		Key word for user field USERT1	Key word displayed when creating / closing inspection points
TEILLOSPFL	CHAR	1		Indicator: assignment of partial batch to an inspection point required	"X" Partial batch confirmation " " No partial batch confirmation
CHARGPFL	CHAR	1		Indicator: batch management required	"X" Batch managed " " No batch management
QUANTITIES	CHAR	1		Confirmation of quantity required	"X" Confirmation of quantity required " " Confirmation of quantity not required
EVALUATION	CHAR	1		Confirmation of a valuation required, else confirmation by QM	"X" Confirmation of evaluation required " " Confirmation of evaluation not required
KOSTL	CHAR	10		Cost center	Cost center
KZKORRTRAN	CHAR	1		Ind.: Correction transmission	Ind.: Correction transmission
PRUEFSTAT	CHAR	1		Status of the inspection	Entries correspond with status of inspection requirements: "A" → FRE "B" → ABG "C" → STO "D" → UNT "E" → SKL "F" → GES In case PRUEFSTAT = "E", additionally the skip lot flag is set to "1", else to "0"
EINHVORG	CHAR	3		Unit of measure for operation	
RUECKMPP	CHAR	1		Indicator: confirmation of inspection point required. This field is currently not supported.	Not used

4.2 Characteristics in “Z2QAIMV000X000” structure

Field name:	T	L	D	Description	Use in HYDRA
SATZART	CHAR	3		Record type	Not used
RUECKMELNR	NUMC	8		Confirmation number for inspection characteristic	Confirmation number
ERFASSART	CHAR	1		Recording type for insp. charact.	“A” Measured value for single unit “B” Code for single unit “C” Valuation for single unit (OK/not OK) “D” Measured value for sample “E” Code for sample “F” Valuation of a sample “G” Measured value for a characteristic “H” Code for a characteristic “I” Valuation of a characteristic “J” Measured value for single unit in a sample “K” Code for single unit in a sample “L” Valuation for single unit in a sample “M” Measured value for sample of an inspection point “N” Code for single unit of an inspection point “O” Valuation for single unit of an inspection point “P” Measured value for sample of an inspection point “Q” Code for sample of an inspection point “R” Valuation for sample of an inspection point
KZBEWSUBSY	CHAR	1		Ind.: valuation by subsystem	“X” Valuation in HYDRA “ ” Valuation in SAP
BEWART	CHAR	1		Valuation type for insp. charact.	“A” Valuation by number of non-conforming units (N-C relation) “B” Valuation by number of defects (N-C relation) “C” Valuation according to s method (ISO3951) “D” Valuation according to code “E” Manual valuation “F” Valuation based on the mean value within tolerance limits “G” Valuation at sample level “H” Valuation according to control chart

Field name:	T	L	D	Description	Use in HYDRA
KZRZWANG	CHAR	1		Results recording required	" " Inspection for characteristic is optional "+" Inspection required if result for preceding required characteristic is OK "-" Inspection required if result for preceding required characteristic is not OK "X" Inspection for characteristic is required
STATUSV	CHAR	1			Not used
STATUSR	CHAR	1			Not used
KZPRUMF	CHAR	1		Ind.: inspection scope	"=" Specified scope of insp. must be adhered to "<" Scope of insp. may be below specification ">" Specified scope of inspection may be exceeded " " Scope of insp. may fall below or exceed specification
KZDOKU	CHAR	1		Documentation required	" " Inspection description for characteristic is optional "." (dot) Inspection description required in case of rejection "+" Inspection description required
KZSERNR	CHAR	1		Ind.: record serial number	" " Serial number optional with single values "X" Serial number required with single values
KZTSTICHPR	CHAR	1		Ind.: partial samples for charact.	" " Inspect single sample for characteristic "X" Inspect multiple sample for characteristic
KZRAST	CHAR	1		Ind.: inspection with insp. grid	" " Sampling procedure without inspection grid "X" Sampling procedure with inspection grid
RASTER	NUMC	3		Inspection frequency within inspection grid	Inspection frequency within inspection grid
SOLLSTPANZ	CHAR	5		No. of partial samples planned	No. of partial samples planned

Field name:	T	L	D	Description	Use in HYDRA
BEWARTSP	CHAR	1		Valuation type of partial sample	"A" Valuation by number of non-conforming units (N-C relation) "B" Valuation by number of defects (N-C relation) "C" Valuation according to s method (ISO3951) "D" Valuation according to code "E" Manual valuation "F" Valuation based on the mean value with- in tolerance limits "G" Valuation at sample level "H" Valuation according to control chart
PRUEFLOS	NUMC	12		Number of the inspection batch	Number of the inspection batch
PLNFL	CHAR	6		Number of the task list sequence	In accordance with configuration (*1)
VORNR	CHAR	4		Operation number	In accordance with configuration (*1)
MERKNR	NUMC	4		Characteristic number	Characteristic number
QPMK_WERKS	CHAR	4		Plant of master inspection charact.	Plant of master inspection charact.
VERWMERKM	CHAR	8		Master inspection characteristic	Master inspection characteristic
MKVERSION	CHAR	6		Version of master insp. charact.	Version of master insp. charact.
QMTB_WERKS	CHAR	4		Plant of the inspection method	Plant of the inspection method
PMETHODE	CHAR	8		Inspection method	Inspection method
PMTVERSION	CHAR	6		Version of the inspection method	Version of the inspection method
PMTKURZTXT	CHAR	40		Short text of the inspection method	Short text of the inspection method
PRUEFQUALI	CHAR	5		Inspector qualification	Inspector qualification
MERKGEW	CHAR	2		Weighting of insp. characteristic	Weighting of insp. characteristic
GEWKURZTXT	CHAR	40		Short text for weighting of inspection characteristic	Short text for weighting of inspection characteristic
KURZTEXT	CHAR	40		Short text of the characteristic	Short text of the characteristic (only for the native MDBI column)
FORMEL	CHAR	120		Formula for inspection charact.	Formula for inspection charact.
DUMMY10	CHAR	10		Additional information 1	Customer / project specific usage
DUMMY20	CHAR	20		Additional information 2	Customer / project specific usage
DUMMY40	CHAR	40		Additional information 3	Customer / project specific usage
STELLEN	NUMC	2		No. of digits after decimal point	No. of digits after decimal point
MASSEINHSW	CHAR	3		Unit of measure for insp. charact.	Unit of measure for insp. charact.
SOLLWERT	CHAR	16		Target value/nominal value	Target value/nominal value
TOLERANZOB	CHAR	16		Upper tolerance limit	Upper tolerance limit
TOLERANZUN	CHAR	16		Lower tolerance limit	Lower tolerance limit
PLAUSIOBEN	CHAR	16		Upper plausibility limit	Upper plausibility limit
PLAUSIUNTE	CHAR	16		Lower plausibility limit	Lower plausibility limit
GRENZEOB1	CHAR	16		First upper limit value	Customer / project specific usage
GRENZEUN1	CHAR	16		First lower limit value	Customer / project specific usage
GRENZEOB2	CHAR	16		Second upper limit value	Customer / project specific usage
GRENZEUN2	CHAR	16		Second lower limit value	Customer / project specific usage

Field name:	T	L	D	Description	Use in HYDRA
KATAB1	CHAR	1		Ind.: catalog entry 1 is selected set	"X" catalog entry 1 is selected set " " catalog entry 1 is not selected set
KATALGART1	CHAR	1		Catalog type 1	Catalog type 1
AUSWMGWRK1	CHAR	4		Plant of selected set 1	Plant of selected set 1
AUSWMENGE1	CHAR	8		Selected set / code group 1	Selected set / code group 1
KATAB2	CHAR	1		Ind.: catalog entry 2 is selected set	"X" catalog entry 2 is selected set " " catalog entry 2 is not selected set
KATALGART2	CHAR	1		Catalog type 2	Catalog type 2
AUSWMGWRK2	CHAR	4		Plant of selected set 2	Plant of selected set 2
AUSWMENGE2	CHAR	8		Selected set / code group 2	Selected set / code group 2
KATAB3	CHAR	1		Ind.: catalog entry 3 is selected set	"X" catalog entry 3 is selected set " " catalog entry 3 is not selected set
KATALGART3	CHAR	1		Catalog type 3	Catalog type 3
AUSWMGWRK3	CHAR	4		Plant of selected set 3	Plant of selected set 3
AUSWMENGE3	CHAR	8		Selected set / code group 3	Selected set / code group 3
KATAB4	CHAR	1		Ind.: catalog entry 4 is selected set	"X" catalog entry 4 is selected set " " catalog entry 4 is not selected set
KATALGART4	CHAR	1		Catalog type 4	Catalog type 4
AUSWMGWRK4	CHAR	4		Plant of selected set 4	Plant of selected set 4
AUSWMENGE4	CHAR	8		Selected set / code group 4	Selected set / code group 4
KATAB5	CHAR	1		Ind.: catalog entry 5 is selected set	"X" catalog entry 5 is selected set " " catalog entry 5 is not selected set
KATALGART5	CHAR	1		Catalog type 5	Catalog type 5
AUSWMGWRK5	CHAR	4		Plant of selected set 5	Plant of selected set 5
AUSWMENGE5	CHAR	8		Selected set / code group 5	Selected set / code group 5
SOLLSTPUMF	NUMC	7		Sample size to be checked per inspection characteristic	If SOLLSTPANZ > 0, then the sample size is calculated through the formula SOLLSTPUMF / SOLLSTPANZ
PROBEMGEH	CHAR	3		Unit of measure for sample	Unit of measure for sample
PROBMGFAK	NUMC	6		Factor for sample unit of measure	Factor for sample unit of measure
ANNAHME	NUMC	5		Acceptance number for attributive inspection	Acceptance number for attributive inspection
RUECKWEZ	NUMC	5		Rejection number for attributive inspection	Rejection number for attributive inspection
KFAKTOR	CHAR	16		K factor for variable inspection	K factor for variable inspection
QRKNR	NUMC	12		Control chart number	Transformation of SAP control chart number into HYDRA control chart number
PHYSPROBE	NUMC	6		Number of the physical sample	Number of the physical sample
KZKORRTRAN	CHAR	1		Ind.: correction transmission	" " First transmission "X" Correction transmission
ZAHL	NUMC	8		Version counter	Version counter

Field name:	T	L	D	Description	Use in HYDRA
ANTVERF	CHAR	1		Share calculation procedure	"A" Binomial distribution "B" Poisson distribution "C" Normal distribution "" Distribution not specified

4.3 Catalog master data in “Z2QAICA000X000” structure

Field name:	T	L	D	Description	Use in HYDRA
SATZART	CHAR	3		Record type	Not used
KATAB	CHAR	1		Ind.: entry is selected set	“X” entry is selected set “X” entry is no selected set
KATALGART	CHAR	1		Catalog type	Catalog type
AUSWMGWRK	CHAR	4		Plant of the selected set	Plant of the selected set
AUSWMENGE	CHAR	8		Selected set	Selected set
CODEGRUPPE	CHAR	8		Code group	Code group
CODE	CHAR	4		Code	Code
KURZTEXT	CHAR	40		Short text of the code	Short text of the code (only for the native MDBI column)
BEWERTUNG	CHAR	1		Valuation	“A” Acceptance (OK) “R” Rejection (not OK) “ ” Valuation not carried out
FEHLKLASSE	CHAR	2		Defect class	Always the same values allowed. Defined in Customizing (transaction OQC7)
MUSSTEXTKZ	CHAR	1		Ind.: text required for confirmation.	“ ” Text not obligatory “X” Text obligatory
BB_VORSCH	CHAR	1		Ind.: carry out inventory posting	“ ” No inventory posting “X” Inventory posting carried out
QKENNZahl	NUMC	3		Quality score	Quality score

4.4 Inspection Points in “Z2QAIPP000X000” structure

Field name:	T	L	D	Description	Use in HYDRA
SATZART	CHAR	3		Record type	“Q85” Download of inspection point
PRUEFLOS	NUMC	12		Inspection batch number	
PLNFL	CHAR	6		Operation sequence in task list	
VORNR	CHAR	4		Operation number	
PROBENR	NUMC	6		Sample number	
TEILLOS	NUMC	6		Partial batch number	
MENGE	CHAR	17		Inspection point quantity	
EINHPR	CHAR	3		Unit of measure for inspection point	Not used
EQUNR	CHAR	18		Equipment number Cannot be defined freely (value range determined by inspection batch); inspection points of type 1 through 3 already defined (these inspection points can be retrieved with function module	
TPLNR	CHAR	30		Number of functional location (see EQUNR)	
PHYNR	CHAR	12		Number of physical sample (see EQUNR)	
USERC1	CHAR	18		User field for 18 characters	
USERC2	CHAR	10		User field for 10 characters	
USERN1	NUMC	10		User field for 10 digits	
USERN2	NUMC	3		User field for 3 digits	
USERD1	DATE	8		User field for date	
USERT1	TIME	6		User field for time	
VKATART	CHAR	1		Catalog type	Not used (only for upload)
VWERKS	CHAR	4		Plant	Not used (only for upload)
VAUSWAHLMG	CHAR	8		Selected set of the usage decision for the inspection point	Not used (only for upload)
VCODEGRP	CHAR	8		Code group of the usage decision	Not used (only for upload)
VCODE	CHAR	4		Code of the usage decision	Not used (only for upload)
VTEXT	CHAR	40		Short text for partial batch	
MATNR	CHAR	18		Material number	Not used
CHARG	CHAR	10		Batch number	
PRUEFDATUM	DATE	8		Start date of the inspection	Not used
PRUEFZEIT	TIME	6		Start time of the inspection	Not used
PRUEFER	CHAR	12		Name of the inspector	Not used
KZRMART	CHAR	1		Confirmation type, currently not used	Not used
URSACHEAS	CHAR	4		Reason for scrap, currently not used	Not used
MENGEAS	CHAR	17		Scrap quantity	
MENGENA	CHAR	17		Rework quantity	

4.5 Error messages in “Z2QIERR000X000” structure

Field name:	T	L	D	Description	Use in HYDRA
LFDNR	NUMC	4		Consecutive number	Usage in HYDRA Escalation Management (Acronym complies with field name)
MSGID	CHAR	20		Message class	Vide field LFDNR
MSGNR	NUMC	3		Message number	Vide field LFDNR
MSGTYPE	CHAR	1		Message type (E, I, W,...)	Vide field LFDNR
MSGTEXT	CHAR	73		Message text	Vide field LFDNR
LOG_NO	CHAR	20		Application log: protocol number	Vide field LFDNR
LOG_MSG_NO	NUMC	6		Application log: number of current message	Vide field LFDNR
PARAM_NAME	CHAR	32		Parameter name	Vide field LFDNR
PARAM_ROW	NUMC	10		Line in parameter	Vide field LFDNR
PARAM_FIELD	CHAR	30		Field in parameter	Vide field LFDNR
PRUEFLOS	NUMC	12		Inspection batch number	Vide field LFDNR
PLNFL	CHAR	6		Sequence of operations within a task list	Vide field LFDNR
VORNR	CHAR	4		Operation number	Vide field LFDNR
VORGLFNR	NUMC	8		Consecutive node number from order counter APLZL	Vide field LFDNR
MERKNR	NUMC	4		Inspection characteristic number	Vide field LFDNR
KATAB	CHAR	1		Indicator: catalog entry is a selected set	Vide field LFDNR
KATALGART	CHAR	1		Catalog type of the assigned code group or selected set	Vide field LFDNR
AUSWMGWRK	CHAR	4		Plant of the assigned selected set	Vide field LFDNR
AUSWMENGE	CHAR	8		Assigned code group or selected set	Vide field LFDNR
CODEGRUPPE	CHAR	8		Code group	Vide field LFDNR
CODE	CHAR	4		Code	Vide field LFDNR
RUECKMELNR	NUMC	8		Confirmation number for the inspection point	Vide field LFDNR
PROBENR	NUMC	6		Number of the sample	Vide field LFDNR
STUECKNR	NUMC	4		Consecutive number for unit to be inspected	Vide field LFDNR
SATZART	CHAR	3		Record types	Vide field LFDNR

5 Upload of inspection results / insp. points / usage decisions

5.1 Upload for Single Results

Message Type:: ZHYQMIDI_ORIGINAL_VALUES

IDoc Type:: ZHYQMIDI_ORIGINAL_VALUES01

Segment:: Z2QAISE000X000 (single results)

Field name:	T	L	D	Description	Use in HYDRA
SATZART	CHAR	3		Record type	"Q51" Quantitative single result "Q52" Code as single result "Q53" Valuation OK/not OK as single result "Q54" Quantitative single result for inspection point "Q55" Code as single result for inspection point "Q56" Valuation as single result for inspection point
RUECKMELNR	NUMC	8		Confirmation no. of the charact.	Confirmation number in accordance with specification
PROBENR	NUMC	6		Number of the partial sample	- Number of partial sample (without inspection points) - Number of inspection point
STUECKNR	NUMC	4		Consecutive number for test units	- Number of sample (without inspection points) - Number of inspection point
KZSERNR	CHAR	1		Ind.: serial number filled	"X" If recording of serial numbers is required " " If recording of serial numbers is not required
SERIALNR	CHAR	18		Serial no. of the test unit	Contains serial number if recording of serial number is required
KZLWERT	CHAR	1		Ind.: last single value	Not used
KZLPROBE	CHAR	1		Ind.: last sample	Not used
KZABSCHL	CHAR	1		Ind.: close characteristic (sample)	Not used
KZBEWEEXT	CHAR	1		Ind.: transfer valuation	Not used
ATTRIBUT	CHAR	1		Attribute of the individual result	"/" Single value was set invalid " " Valid value
MESSWERT	CHAR	16		Measured value	Measured value
GRUPPE1	CHAR	8		Code group 1	Code group 1
CODE1	CHAR	4		Code 1	Code 1
GRUPPE2	CHAR	8		Code group 2	Code group 2
CODE2	CHAR	4		Code 2	Code 2
GRUPPE3	CHAR	8		Code group 3	Code group 3
CODE3	CHAR	4		Code 3	Code 3
GRUPPE4	CHAR	8		Code group 4	Code group 4
CODE4	CHAR	4		Code 4	Code 4

Field name:	T	L	D	Description	Use in HYDRA
GRUPPE5	CHAR	8		Code group 5	Code group 5
CODE5	CHAR	4		Code 5	Code 5
BEWERTUNG	CHAR	1		Valuation	"A" Acceptance (OK) "R" Rejection (not OK)
FEHLKLAS	CHAR	2		Defect class	Defect class if available
ANZFEHLER	NUMC	2		Number of defects	Number of defects
PRUEFDATUV	DATE	8		Start date of the inspection	Start date of the inspection
PRUEFZEITV	TIME	6		Start time of the inspection	Start time of the inspection
PRUEFER	CHAR	12		Name of the inspector	If result was recorded at HYDRA console, it contains the user; otherwise it contains the personal card number.
QERGDATH	CHAR	2		Origin of results data (on completion)	According to customizing
MASCHINE	CHAR	18		Machine	Machine / work centre
POSITION	NUMC	4		Position on the machine	Position
PRUEFBEMKT	CHAR	40		Short text for inspection description	Short text for inspection description
MBEWERTGPR	CHAR	1		Valuation of the sample	Not used
FEHLKLASPR	CHAR	2		Defect class for sample valuation	Not used
MBEWERTGMK	CHAR	1		Valuation of the characteristic	Not used
FEHLKLASMK	CHAR	2		Defect class for characteristic valuation	Not used

5.2 Upload for Sample Results

Message type: ZHYQMIDI_SAMPLE_VALUES
IDoc type: ZHYQMIDI_SAMPLE_VALUES01
Segment:: Z2QAISR000X000 (sample results)

Field name:	T	L	D	Description	Use in HYDRA
SATZART	CHAR	3		Record type	"Q61" Quantitative sample result "Q62" Code as sample result "Q63" Valuation OK/not OK for sample "Q64" Quantitative result for inspection point "Q65" Code for inspection point "Q66" Valuation OK/not OK for inspection point "Q69" Close sample record
RUECKMELNR	NUMC	8		Confirmation no. of the charact.	Confirmation number in accordance with specification
PROBENR	NUMC	6		Ind.: close characteristic	- Number of partial sample (without inspection points) - Number of inspection point
KZLRPOBE	CHAR	1		Ind.: transfer valuation	Not used
KZABSCHL	CHAR	1		Attribute of the results record	Not used
KZBEWEEEXT	CHAR	1		Valuation of characteristic for usage decision	Not used
ATTRIBUT	CHAR	1		Defect class	"/" Single value was set invalid "" Valid value
GRUPPE1	CHAR	8		Code group 1	Code group 1
CODE1	CHAR	4		Code 1	Code 1
GRUPPE2	CHAR	8		Code group 2	Code group 2
CODE2	CHAR	4		Code 2	Code 2
GRUPPE3	CHAR	8		Code group 3	Code group 3
CODE3	CHAR	4		Code 3	Code 3
GRUPPE4	CHAR	8		Code group 4	Code group 4
CODE4	CHAR	4		Code 4	Code 4
GRUPPE5	CHAR	8		Code group 5	Code group 5
CODE5	CHAR	4		Code 5	Code 5
ANZWERTG	NUMC	4		Number of valid values	Number of valid values if QAIMV.BEWARTSP = "A" or "B"
ANZFEHLEH	NUMC	4		Number of non-conforming units	Number of non-conforming units if QAIMV.BEWARTSP = "A"
ANZFEHLER	NUMC	4		Number of defects	Number of defects if QAIMV.BEWARTSP = "B"
ANZWERTO	NUMC	4		Values above upper tolerance limit	Values above upper tolerance limit
ANZWERTU	NUMC	4		Values below lower tolerance limit	Values below lower tolerance limit

Field name:	T	L	D	Description	Use in HYDRA
MITTELWERT	CHAR	16		Average of valid measured values	Average of valid measured values if QAIMV.BEWARTSP = "C"
VARIANZ	CHAR	16		Variance of valid measured values	Average of valid measured values if QAIMV.BEWARTSP = "C"
MAXWERT	CHAR	16		Max. of valid measured values	Max. of valid measured values
MEDIANWERT	CHAR	16		Median of valid measured values	Median of valid measured values
MINWERT	CHAR	16		Min. of valid measured values	Min. of valid measured values
PRUEFDATUV	DATE	8		Start date of the inspection	Start date of the inspection
PRUEFDATUB	DATE	8		Finish date of the inspection	Finish date of the inspection
PRUEFZEITV	TIME	6		Start time of the inspection	Start time of the inspection
PRUEFZEITB	TIME	6		Finish time of the inspection	Finish time of the inspection
PRUEFER	CHAR	12		Name of the inspector	If result was recorded at HYDRA console, it contains the user; otherwise it contains the personal card number.
QERGDATH	CHAR	2		Origin of results data	According to customizing
MASCHINE	CHAR	18		Machine	Machine / work centre
POSITION	NUMC	4		Position on the machine	Position
PRUEFBEMKT	CHAR	40		Short text for inspection descript.	Short text for inspection description
MBEWERTGPR	CHAR	1			Not used
FEHLKLASPR	CHAR	2			Not used
MBEWERTGMK	CHAR	1			Not used
FEHLKLASMK	CHAR	2			Not used

5.3 Upload for Characteristic Results

Message type: ZHYQMIDI_FEATURE_VALUES
IDoc type: ZHYQMIDI_FEATURE_VALUES01
Segment: Z2QAIMR000X000 (characteristic results)

Field name:	T	L	D	Description	Use in HYDRA
SATZART	CHAR	3		Record type	"Q71" Quantitative characteristic result "Q72" Code as characteristic result "Q73" Valuation OK/not OK for characteristic "Q79" Close characteristic
RUECKMELNR	NUMC	8		Confirmation no. of the charact.	Confirmation number in accordance with specification
KZABSCHL	CHAR	1		Ind.: close characteristic	"X" If record type "Q79" " " All other record types
KZBEWEEXT	CHAR	1		Ind.: transfer valuation	Ind.: transfer valuation in accordance to specification
ATTRIBUT	CHAR	1		Attribute of the results record	"/" Single value was set invalid " " Valid value
MBEWERTG	CHAR	1		Valuation of characteristic for usage decision	"A" Acceptance (OK) "R" Rejection (not OK)
FEHLKLAS	CHAR	2		Defect class	Defect class if available
GRUPPE1	CHAR	8		Code group 1	Code group 1
CODE1	CHAR	4		Code 1	Code 1
GRUPPE2	CHAR	8		Code group 2	Code group 2
CODE2	CHAR	4		Code 2	Code 2
GRUPPE3	CHAR	8		Code group 3	Code group 3
CODE3	CHAR	4		Code 3	Code 3
GRUPPE4	CHAR	8		Code group 4	Code group 4
CODE4	CHAR	4		Code 4	Code 4
GRUPPE5	CHAR	8		Code group 5	Code group 5
CODE5	CHAR	4		Code 5	Code 5
ANZWERTG	NUMC	7		Number of valid values	Number of valid values if QAIMV.BEWARTSP = "A" or "B"
ANZFEHLEH	NUMC	7		Number of non-conforming units	Number of non-conforming units if QAIMV.BEWARTSP = "A"
ANZFEHLER	NUMC	7		Number of defects	Number of defects if QAIMV.BEWARTSP = "B"
ANZWERTO	NUMC	7		Values above upper tolerance limit	Values above upper tolerance limit
ANZWERTU	NUMC	7		Values below lower tolerance limit	Values below lower tolerance limit
MITTELWERT	CHAR	16		Average of valid measured values	Average of valid measured values if QAIMV.BEWARTSP = "C"
VARIANZ	CHAR	16		Variance of valid measured values	Average of valid measured values if QAIMV.BEWARTSP = "C"

Field name:	T	L	D	Description	Use in HYDRA
MAXWERT	CHAR	16		Max. of valid measured values	Max. of valid measured values
MEDIANWERT	CHAR	16		Median of valid measured values	Median of valid measured values
MINWERT	CHAR	16		Min. of valid measured values	Min. of valid measured values
IVARIANZ	CHAR	16		Inner variance of measured values	Inner variance of measured values
PRUEFDATUV	DATE	8		Start date of the inspection	Start date of the inspection
PRUEFDATUB	DATE	8		Finish date of the inspection	Finish date of the inspection
PRUEFZEITV	TIME	6		Start time of the inspection	Start time of the inspection
PRUEFZEITB	TIME	6		Finish time of the inspection	Finish time of the inspection
PRUEFER	CHAR	12		Name of the inspector	If result was recorded at HYDRA console, it contains the user; otherwise it contains the personal card number.
QERGDATH	CHAR	2		Origin of results data	According to customizing
MASCHINE	CHAR	18		Machine	Machine / work centre
POSITION	NUMC	4		Position on the machine	Position
PRUEFBEMKT	CHAR	40		Short text for inspection description	Short text for inspection description

5.4 Upload for Inspection Points

Message type: ZHYQMIDI_INSP_POINTS
IDoc type:: ZHYQMIDI_INSP_POINTS01
Segment: Z2QAIPP000X000 (inspection points)

Field name:	T	L	D	Description	Use in HYDRA
SATZART	CHAR	3		Record type	"Q83" Creation / update of an inspection point "Q84" Valuation for inspection point
PRUEFLOS	NUMC	12		Inspection batch number	Inspection batch number in accordance to specification
PLNFL	CHAR	6		Operation sequence in task list	Operation sequence in task list in accordance to specification
VORNR	CHAR	4		Operation number	Operation number in accordance to specification
PROBENR	NUMC	6		Sample number	Consecutive number for the inspection point
TEILLOS	NUMC	6		Partial batch number	Recorded value if QAIVC.TEILLOSPFL = „X“
MENGE	CHAR	17		Inspection point quantity	Recorded quantity if QAIVC.QUANTITIES = „X“
EINHPR	CHAR	3		Unit of measure for inspection point	Unit of measure for inspection point
EQUNR	CHAR	18		Equipment number Cannot be defined freely (value range determined by inspection batch); inspection points of type 1 through 3 already defined (these inspection points can be retrieved with function module)	Not used
TPLNR	CHAR	30		Number of functional location (see EQUNR)	Not used
PHYNR	CHAR	12		Number of physical sample (see EQUNR)	Not used
USERC1	CHAR	18		User field for 18 characters	Recorded value if QAIVC.KZUSERC1 = „X“ or “1” to “6”
USERC2	CHAR	10		User field for 10 characters	Recorded value if QAIVC.KZUSERC2 = „X“ or “1” to “6”
USERN1	NUMC	10		User field for 10 digits	Recorded value if QAIVC.KZUSERN1 = „X“ or “1” to “6”
USERN2	NUMC	3		User field for 3 digits	Recorded value if QAIVC.KZUSERN2 = „X“ or “1” to “6”
USERD1	DATE	8		User field for date	Recorded value if QAIVC.KZUSERD1 = „X“ or “1” to “6”
USERT1	TIME	6		User field for time	Recorded value if QAIVC.KZUSERT1 = „X“ or “1” to “6”

Field name:	T	L	D	Description	Use in HYDRA
VKATART	CHAR	1		Catalog type	"Q83" Not used "Q84" Catalog type
VWERKS	CHAR	4		Plant	"Q83" Not used "Q84" Plant
VAUSWAHLMG	CHAR	8		Selected set of the usage decision for the inspection point	"Q83" Not used "Q84" Selected set of the usage decision for the inspection point
VCODEGRP	CHAR	8		Code group of the usage decision	"Q83" Not used "Q84" Code group of the usage decision
VCODE	CHAR	4		Code of the usage decision	"Q83" Not used "Q84" Code of the usage decision
VTEXT	CHAR	40		Short text for partial batch	"Q83" Not used "Q84" Recorded value
MATNR	CHAR	18		Material number	Material number in accordance to specification
CHARG	CHAR	10		Batch number	Recorded value if QAIVC.CHARGPFL = "X"
PRUEFDATUM	DATE	8		Start date of the inspection	Start date of the inspection
PRUEFZEIT	TIME	6		Start time of the inspection	Start time of the inspection
PRUEFER	CHAR	12		Name of the inspector	If result was recorded at HYDRA console, it contains the user; otherwise it contains the personal card number.
KZRMART	CHAR	1		Confirmation type, currently not used	Not used
URSACHEAS	CHAR	4		Reason for scrap, currently not used	Not used
MENGEAS	CHAR	17		Scrap quantity	"Q83" Not used "Q84" Recorded value
MENGENA	CHAR	17		Rework quantity	"Q83" Not used "Q84" Recorded value

5.5 Upload for Usage Decision

Message type: ZHYQMIDI_USAGE_DECISION
IDoc type: ZHYQMIDI_USAGE_DECISION01
Segment: Z2QAIVE000X000 (usage decision)

Field name:	T	L	D	Description	Use in HYDRA
SATZART	CHAR	3		Record type	"Q88" Transfer of usage decision "Q89" Cancellation of inspection, usage decision transferred

Field name:	T	L	D	Description	Use in HYDRA
PRUEFLOS	NUMC	12		Inspection batch number	Inspection batch number in accordance to specification
AUSWMENGE	CHAR	8		Selected set of the usage decision	Selected set of the usage decision in accordance to specification
AUSWMGWRK	CHAR	4		Plant of the selected set	Plant of the selected set in accordance to specification
CODE	CHAR	4		Code	Recorded Value
CODEGRUPPE	CHAR	8		Code group	Recorded Value
VNAME	CHAR	12		Person who made usage decision	If result was recorded at HYDRA console, it contains the user; otherwise it contains the personal card number.
VDATUM	DATE	8		Date when usage decision was made	Date when usage decision was made
VZEIT	TIME	6		Time when usage decision was made	Time when usage decision was made
VTEXT	CHAR	80		Text for usage decision	Text for usage decision

5.6 Upload for Defect Items

Message type: ZHYQMIDI_DEFECT_ITEMS
IDoc type: ZHYQMIDI_DEFECT_ITEMS 01
Segment: Z2QMIFE000X000 (defect items)

Field name:	T	L	D	Description	Use in HYDRA
SATZART	CHAR	3		Record type	"Q90" Defect item for in- spection batch "Q91" Defect item for operation "Q92" Defect item for characteristic in- dependent multiple sample "Q95" Defect item for operation with reference to in- spection point "Q96" Defect item for characteristic with ref- erence to inspection point
PRUEFLOS	NUMC	12		Inspection batch number	Inspection point number in accord- ance to specification
PLNFL	CHAR	6		Sequence of operations in task list	Sequence of operations in task list in accordance to specification
VORNR	CHAR	4		Operation number	Operation number in accordance to specification
MERKNR	NUMC	4		Characteristic number	Characteristic number in accord- ance to specification
PROBENR	NUMC	6		Number of partial sample/inspection point	Number of partial sam- ple/inspection point
RUECKMELNR	NUMC	8		Confirmation number for inspection characteristic	Confirmation number for inspection characteristic in accordance to specification
POSNR	NUMC	4		Sort number for item	Consecutive number
FEKAT	CHAR	1		Catalog type - defects	Catalog type - defects
FEGRP	CHAR	8		Code group - defects	Code group - defects
FECOD	CHAR	4		Defects	Defects
SERIALNR	CHAR	18		Single-unit number of unit to be in- spected	Single-unit number of unit to be in- spected
ANZFEHLER	CHAR	7		Number of defects	Number of defects
FEQLAS	CHAR	2		Defect class	Defect class
KZSYSFE	CHAR	1		Indicator: systematic defect	Indicator: systematic defect
OTKAT	CHAR	1		Catalog type - object parts	Catalog type - object parts
OTGRP	CHAR	8		Code group - object parts	Code group - object parts
OTEIL	CHAR	4		Object part	Object part
FETXT	CHAR	40		Short text for defect item	Short text for defect item
BAUTL	CHAR	18		Assembly	Assembly
FEHLBEW	CHAR	10		Quantitative defect valuation	Quantitative defect valuation
UNITFLBEW	UNIT	3		Unit for defect valuation	Unit for defect valuation

Field name:	T	L	D	Description	Use in HYDRA
FENAM	CHAR	12		Name of person who processed defect record	If result was recorded at HYDRA console, it contains the user; otherwise it contains the personal card number.
FEDAT	DATS	8		Date of record processing	Date of record processing
FZEIT	TIMS	6		Time of record processing	Time of record processing

6 Application-relevant settings in HYDRA

6.1 Configuring variants for Stand-alone CAQ

The request for inspection specifications is based on a uniform selection table. Opposite to the integration scenarios with PP-PDC and/or HYINFO, for stand-alone CAQ integration it is not possible to use any data provided by SAP to request the specifications.

For the following reason there is a customizing table in HYDRA. The table represents the QAILS selection structure defined by SAP and allows though, requesting inspection specifications from SAP. Right now there is no graphical user interface to maintain the table, though any adjustments have to be done by using database interface.

Table: **CAQ_QAILS_VORMERK**

Field	T	L	Description	Meaning / Remark
SATZART	CHAR	3	Record type for request record	Fixed „Q40“
LOSNR_VON	NUMC	12	From inspection batch number	Should not be used for selection
LOSNR_BIS	NUMC	12	To inspection batch number	Should not be used for selection
PLNFL	CHAR	6	Operation sequence in task list	Should not be used for selection
VORNR_VON	CHAR	4	From operation number	Should not be used for selection
VORNR_BIS	CHAR	4	To operation number	Should not be used for selection
VORGWERK	CHAR	4	Plant of operation to be processed	Should be used for selection
SUBSYS	CHAR	6	Identifier of the subsystem	Value as defined in SAP customizing (please see also chapter 7.1 <i>Defining QM subsystem</i>) Default: QM0001
PRPLATZ	CHAR	8	Work center	Useable for selection
PRPLATZWRK	CHAR	4	Plant of the work center	Useable for selection
MATNR	CHAR	18	Material number	Useable for selection
DATUM_VON	CHAR	15	From creation date of inspection batch	It is possible to calculate the “date from” dynamically according to the current date minus x days. For that the entry has to be done such as: TODAY-n (n represents the number of days, e.g. 5) Default: TODAY
DATUM_BIS	CHAR	15	To creation date of inspection batch	It is possible to calculate the “date from” dynamically according to the current date minus x days. For that the entry has to be done such as: TODAY-n (n represents the number of days, e.g. 5) Default: Today
PRUEFSTAT]	CHAR	1	Status of the inspection	Useable for selection

Field	T	L	Description	Meaning / Remark
ART	CHAR	8	Inspection type	Useable for selection
HERKUNFT	CHAR	2	Origin of the inspection batch	Useable for selection
CHARG	CHAR	10	Batch number	Should not be used for selection
AUFNR_VON	CHAR	12	From order number	Should not be used for selection
AUFNR_BIS	CHAR	12	To order number	Should not be used for selection
LIFNR	CHAR	10	Vendor number	Should not be used for selection
KUNNR	CHAR	10	Customer number	Useable for selection
MBLNR	CHAR	10	Number of the material document	Should not be used for selection
MAXLOSANZ	NUMC	4	Maximum number of batches per transmission	Fixed "9999"
BEARB	CHAR		User	Future use
TA_ID	CHAR	30		Future Use
STATISCH	CHAR	1	Flag "Static entry"	Fixed "J"
STATUS	CHAR	3		Future use
BEARB_DATE	DATE			Future use
BEARB_TIME	TIME			Future use
USER_D_01	DATE		User field	Future use
USER_D_02	DATE		User field	Future use
USER_N_03	NUMC	8	User field	Future use
USER_N_04	NUMC	8	User field	Future use
USER_N_05	NUMC	8	User field	Future use
USER_F_07	DEC	13,3	User field	Future use
USER_F_08	DEC	13,3	User field	Future use
USER_C_09	CHAR	20	User field	Future use
USER_C_10	CHAR	40	User field	Future use
Verweis			Database serial	Consecutive number

The hysapqmc.exe/out program requests the inspection specifications. In the stand-alone CAQ integration the program has to be used with the following parameters:

/MESTYP_OUT=

The "/MESTYP_OUT" program parameter defines the message type generated for inspection specifications. For the stand-alone CAQ integration the "ZQM_IDI" value has to be used.

/LOGSYS=

The "/LOGSYS" program parameter corresponds to the customizing setting for the logical system in HYDRA MES Link Enabling (MLE).

/VARIANTE=

The "/VARIANTE" program parameter refers to a variant in table SAP_FB_PARAM_CFG for function module QIRF_SEND_REQUIREMENTS_GET_DAT2.

To schedule the request for inspection specifications on a cyclic base an entry in the HYDRA Scheduler has to be done:

Model: I

Type C

Product key: HKMQM-IDI

License key: HKMQM-IDI

Command:

```
. sh.exe hysapqmc.scr /MESTYP_OUT=ZQM_IDI /LOGSYS=<logical system>  
/VARIANTE=QM_IDI
```

NOTE:

For UNIX systems the entry in the Command field has to look like:

```
./hysapqmc.scr /MESTYP_OUT=ZQM_IDI /LOGSYS=<logical system> /VARIANTE=QM_IDI
```

6.2 Configuring function module parameters

The integration of HYDRA with SAP QM by using QM-IDI requires the use of a number of function modules in SAP. Each function module provides a set of import parameters that allow controlling the selection or posting process in SAP.

In order to provide the highest flexibility to control these ones differently, all these parameters can be maintained in a customizing table in HYDRA. Right now there is no graphical user interface to maintain the table, though any adjustments have to be done by using database interface.

Table: **SAP_FB_PARAM_CFG**

Field	T	L	Description	Meaning / Remark
VARIANTE	CHAR	30	Variant	The variant is the logical name for a set of parameters for one or more function modules.
FB_NAME	CHAR	30	Name of the function module	Together with the field VARIANTE the FB-NAME forms the key of the table. The field contains the technical name as it can be found in SAP transaction SE37 or as it is described in chapter 3 <i>Mapping the QM-IDI in HYDRA</i> , e.g. QIRF_SEND_REQUIREMENTS_GET_DAT2
PARAM_NAME	CHAR	50	Name of the parameter	The field contains the technical name as it can be found in SAP transaction SE37 or as it is described in chapter 3 <i>Mapping the QM-IDI in HYDRA</i> , e.g. I_IND_SORT_ASCENDING_TO_DATE
PARAM_VALUE	CHAR	100	Value of the parameter	Contains the actual value of the parameter, e.g. "X".
PARAM_TYPE	CHAR	15	Type of the parameter	Future use
PARAM_LENGTH	NUM	10	Length of the parameters	Future use
VERWEIS			Database serial	Consecutive number

A variant for each function module used and supported in HYDRA QM-IDI implementation (variant "QM_IDI") is included in the scope of delivery. For each function module the full range of parameters is provided with the default setting as described in chapter 3 *Mapping the QM-IDI in HYDRA*.

6.3 Configuring call for catalog master data

Catalog master data of SAP is needed for all integration scenarios. The catalog master data download is also executed by the hysapqmc.exe/out program. To download catalog master data from SAP the program has to be used with the following parameters:

/MESTYP_OUT=

The "/MESTYP_OUT" program parameter defines the message type generated for catalog master data. For that, the "ZHYQMIDI_CATALOG" value has to be used.

/LOGSYS=

The “/LOGSYS” program parameter corresponds to the customizing setting for the logical system in HYDRA MES Link Enabling (MLE).

/VARIANTE=

The “/VARIANTE” program parameter refers to a variant in the SAP_FB_PARAM_CFG table for the QIRF_SEND_CATALOG_DATA2 function module.

The HYDRA scope of delivery includes the “QM_IDI” variant for the QIRF_SEND_CATALOG_DATA2 function module as an example. In the example all parameter values are empty.

If necessary multiple entries have to be created to download all necessary catalog master data from SAP. It is strictly recommended to create the variants in the customer name space, starting with “U_”.

To schedule the download of catalog master data on a cyclic base an entry in the HYDRA Scheduler has to be done:

Model: I

Type C

Product key: HKMQM-IDI

License key: HKMQM-IDI

Command:

```
. sh.exe hysapqmc.scr /MESTYP_OUT=ZHYQMIDI_CATALOG /LOGSYS=<logical system>  
/VARIANTE=QM_IDI
```

NOTE:

For UNIX systems the entry in the Command field has to look like:

```
./hysapqmc.scr /MESTYP_OUT= ZHYQMIDI_CATALOG /LOGSYS=SAP /VARIANTE=QM_IDI
```

6.4 Configuring call for inspection point download

Inspection points from SAP are needed for all integration scenarios. The inspection point download is also executed by the hysapqmc.exe/out program. To download inspection points from SAP the program has to be used with the following parameters:

/MESTYP_OUT=

The “/MESTYP_OUT” program parameter defines the message type generated for catalog master data. For that, the “ZHYQMIDI_INSPPOINT” value has to be used.

/LOGSYS=

The “/LOGSYS” program parameter corresponds to the customizing setting for the logical system in HYDRA MES Link Enabling (MLE).

/VARIANTE=

The “/VARIANTE” program parameter refers to a variant in the SAP_FB_PARAM_CFG table for the QIRF_INSPPOINT_GETLIST function module.

The HYDRA scope of delivery includes the “QM_IDI” variant for the QIRF_INSPPOINT_GETLIST function module as an example. In the example all parameter values are empty.

If necessary multiple entries have to be created to download all necessary inspection points from SAP. It is strictly recommended to create the variants in the customer name space, starting with “U_”.

To schedule the download of catalog master data on a cyclic base an entry in the HYDRA Scheduler has to be done:

Model: I
Type C
Product key: HKMQM-IDI
License key: HKMQM-IDI
Command:
.
 sh.exe hysapqmc.scr /MESTYP_OUT=ZHYQMIDI_INSPPOINT /LOGSYS=<logical system>
 /VARIANTE=QM_IDI

NOTE:

For UNIX systems the entry in the Command field has to look like:

```
./hysapqmc.scr /MESTYP_OUT= ZHYQMIDI_INSPPOINT /LOGSYS=SAP /VARIANTE=QM_IDI
```

6.5 Configuring for inspection point creation in HYDRA

When inspection points are not downloaded from SAP, they are created in HYDRA. The creation for time and quantity related inspection points is done by a cyclic process, that is scheduled in the HYDRA Scheduler. There the following entry has to be created:

Model: I
Interval 5 minutes
Type C
Product key: QMS-BP
License key: QMS-BP
Command:
. sh.exe hyqmsipcr.scr

NOTE:

For UNIX systems the entry in the Command field has to look like:

./hyqmsipcr.scr

6.6 Configuration of confirmations

After data recording in HYDRA the confirmation data has to be transferred back to SAP. The confirmation is done on a cyclic base (Default: 15 minutes).

To schedule the upload recorded data to SAP on a cyclic base an entry in the HYDRA Scheduler has to be done:

Model: I
Type C
Product key: HKMQM-IDI
License key: HKMQM-IDI
Command:
. sh.exe ./qm_idi_rck.scr

NOTE:

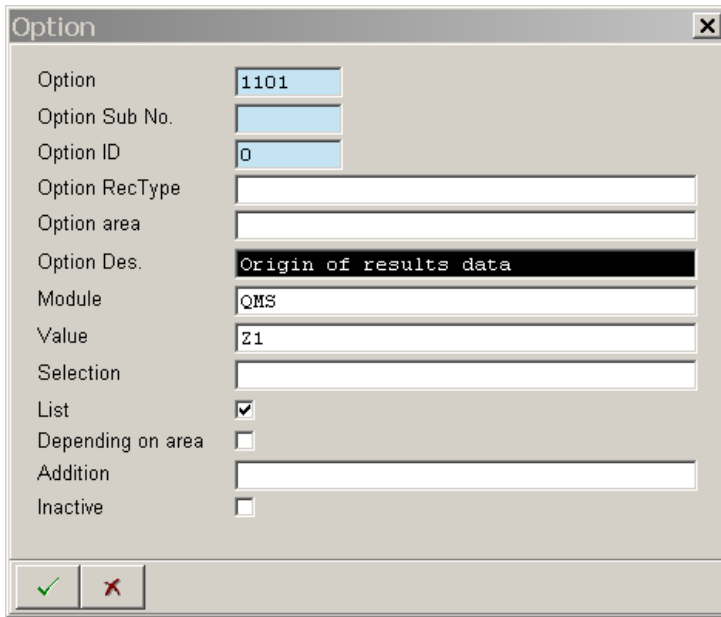
For UNIX systems the entry in the Command field has to look like:

./qm_idi_rck.scr

6.7 Configuring origin of results data

The origin of results data can be configured in SAP customizing. To enable HYDRA to transfer this information when uploading inspection results to SAP, the entry has to be customized in HYDRA accordingly. The customizing is only possible in HYDRA professional mode.

CUS → CAQ → Master data → System Administration → Options



The image shows a dialog box titled "Option" with a close button (X) in the top right corner. It contains several input fields and checkboxes:

- Option: 1101
- Option Sub No.: (empty)
- Option ID: 0
- Option RecType: (empty)
- Option area: (empty)
- Option Des.: Origin of results data
- Module: QMS
- Value: Z1
- Selection: (empty)
- List: ☒
- Depending on area: ☐
- Addition: (empty)
- Inactive: ☐

At the bottom left, there are two buttons: a green checkmark and a red X.

6.8 Configuring partial confirmations for inspection lots

Partial confirmations for inspection lots can be configured HYDRA CAQ customizing.

By activating the CAQ option 1128 (value = Y) partial confirmations for inspection lots will be transferred to SAP even when the final usage decision is not done yet.

Remarks:

By activating the option WITHOUT the additional setting [DIRECT] the system will behave such as:

- Inspection points will be confirmed after closing them. Together with the inspection points single results, characteristic results and defect items (Q95 and Q96) will be transferred.
- For operations that are not inspection point relevant, the characteristic results with their single and sample results will be confirmed when closing the inspection order (usually when finishing the operation).

ATTENTION!! Setting the option might cause problems when reactivating CAQ-relevant operations.

- Defect items for operations (Q91) and characteristics/ independent multiple samples (Q92) will be confirmed when closing the inspection order (usually when finishing the operation).

ATTENTION!! Setting the option might cause problems when reactivating CAQ-relevant operations.

- Defect items of the inspection requirement (Q90) will be confirmed after closing the inspection requirement.
- The usage decision will be confirmed when it is done, usually after closing the inspection requirement.

By activating the option WITH the additional setting [DIRECT] the system will behave such as:

- All inspection point details, recorded results und defect items will be confirmed directly after recording.
- The characteristics will be confirmed as closed, when the assigned inspection order is finished, usually when finishing the last operation.
ATTENTION!! Setting the option might cause problems when reactivating CAQ-relevant operations.
- Sample results will be confirmed as closed, when the assigned inspection order has been closed or the inspection point is closed.
ATTENTION!! Setting the option might cause problems when reactivating CAQ-relevant operations.

The usage decision will be confirmed after recording, usually when the assigned inspection requirement is closed.

ATTENTION!! Activating the option with additional setting [DIRECT] is only available for testing reasons. The setting is NOT released for customers.

In case the option is not available or inactive, all inspection lot data will be transferred after making the final usage decision.

CUS → CAQ → Master data → System Administration → Options

Option	
Option	1128
Option sub no.	
Option ID	0
Option RecType	
Option area	
Option Des.	QM-IDI Rückmeldungen bei Teilabschlü
Module	QMS
Value	N
Selection	Y/N
List	☐
Depending on area	☐
Addition	[DIRECT]
Inactive	☐

6.9 Settings in HYDRA MES Link Enabling

The integration of HYDRA with SAP QM by using QM-IDI requires settings in the HYDRA MES Link Enabling (MLE). The technical communication can be configured within the Logical Systems of MLE. The inbound and outbound processing has to be configured in the HYDRA MLE distribution model.

Detailed information on HYDRA MES Link Enabling can be found in the HYDRA documentation *HKM-MLEK.pdf*.

6.10 Assignment of key values for production orders

In PP-PDC and HYINFO integration scenario general settings for operations from SAP such as the assignment of key values have also to be done.

Detailed information on these settings can be found in the HYDRA documentation *HKMPP-PDC.pdf*.

7 Application-relevant customizing in SAP

7.1 Defining QM subsystem

In QM-IDI interface operations are only transferred if a subsystem is assigned to them. The assignment is done at the work centre in SAP. Each work centre can be assigned to one subsystem only, whereas a subsystem can be assigned to multiple work centers.

QM subsystems can be created in SAP using by IMG.

7.2 Defining origin of results data

In SAP QM it is possible to indicate the origin at the inspection result. For that, several origins can be defined in the IMG. To enable HYDRA to support this field/information it is necessary to maintain the value created in SAP in HYDRA as well (please see chapter 6.7 *Configuring origin of results data* for more details).

7.3 Defining detail level for error messages

When defining QM subsystems, the trace level can be defined. In the IDI interface, all error messages as well as changes to the worklist are written to an application log.

The exceptions, messages of the QIERRTAB error log, and the beginning and end of a function are recorded. In Customizing, you can define the level of detail for the application log. Use the RQEIFML1 report to display the application log. Use the RQEIFML2 report to delete the log.